Slide 1: Title Slide

Title: Spam Email Detection Using Machine Learning

Subtitle: Mini Project
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Slide 2: Introduction

- Email spam is unwanted or harmful messages sent in bulk.
- It can contain advertisements, phishing links, or malware.
- Detecting spam emails helps improve security and user experience.
- Machine Learning can automatically classify emails as spam or not spam.

Slide 3: Objective

- To build a model that classifies emails into:
  - Spam (1)
  - Not Spam / Ham (0)
- Use Machine Learning techniques for text classification.
- Provide real-time prediction for user input emails.

Slide 4: Technologies Used

- Python

- Pandas (Data handling)
- Scikit-learn (ML model)
- Matplotlib (Visualization)
- CountVectorizer (Text to Numeric)

Slide 5: Dataset

- Dataset Name: Emails_data.csv
- Contains two columns:
    - Message (Email text)
    - Label (spam / ham)
- Label conversion:
    - Spam → 1
    - Ham → 0

Slide 6: Data Preprocessing

- Convert labels into numeric values.
- Use CountVectorizer to convert text into numerical format.
- Each word is converted into frequency values.

Slide 7: Feature Extraction

- CountVectorizer is used to transform text data.
- Converts email text into a matrix of token counts.
- Example:
    - "Free offer" → [1, 1, 0, ...]

Slide 8: Data Visualization

- Scatter plot used for visualization.
- X-axis: Word frequency
- Y-axis: Label (0 = Ham, 1 = Spam)
- Helps understand data distribution.

Slide 9: Train-Test Split

- Dataset divided into:
  - Training Data (80%)
  - Testing Data (20%)
- Ensures model is evaluated on unseen data.

Slide 10: Model Used

- Algorithm: Multinomial Naive Bayes • Suitable for text classification problems.
- Works based on probability.

Slide 11: Model Training

- Train model using training data.
- Model learns patterns of spam and non-spam emails.

Slide 12: Prediction

- Model predicts labels for test data.
- Output: Spam or Not Spam

Slide 13: Accuracy Evaluation

- Accuracy Score is used to measure performance.
- Formula:
  Accuracy = Correct Predictions / Total Predictions

Slide 14: User Input Testing

- User can enter any email text.

- Model predicts:
    - Spam Email
    - Not Spam Email
- Loop continues until user types "exit".

Slide 15: Advantages

- Fast and automatic detection
- Reduces manual effort
- Improves email security

Slide 16: Limitations

- Accuracy depends on dataset quality
- Cannot detect new types of spam easily
- Needs proper training data

Slide 17: Future Scope

- Use advanced models like Deep Learning
- Improve accuracy with large datasets
- Add GUI for better user experience

Slide 18: Conclusion

- Successfully built a spam detection system
- Used Machine Learning for classification
- Model can predict spam emails effectively

Slide 19: Thank You Thank
You!
Any Questions?